

Design for class “PayOrderController”

Inheritance of class “BaseController”

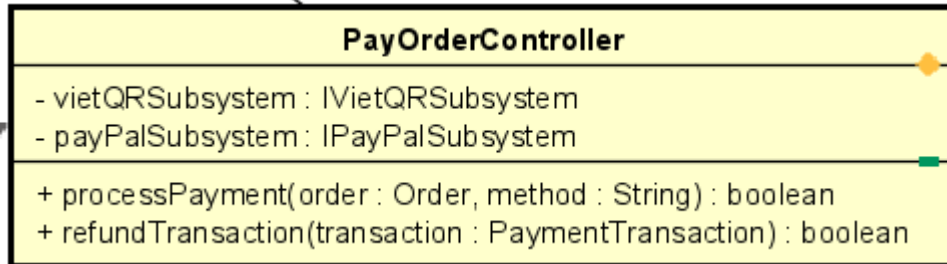


Table 1. Attribute Design

#	Name	Data type	Default value	Description
1	vietQRSubsystem	IVietQRSubsystem	null	Subsystem used for generating and checking VietQR payment.
2	paypalSubsystem	IPayPalSubsystem	null	Subsystem for PayPal payment processing.

Table 2. Operation Design

#	Name	Return type	Description
1	processPayment(order, method)	boolean	Initiate payment using the selected method and return success or failure.
2	refundTransaction(transaction)	boolean	Perform a refund for a given transaction.

Method Design:

1. processPayment

- Parameter:

- order: Order - the order being paid.
- method: String - "QR" or "PAYPAL".

- Exception:

- PaymentFailureException if subsystem reports payment error.

- `UnsupportedPaymentMethodException` if method is unknown.

- Method:

- If method == "QR" → call `vietQRSubsystem.createPayment(order)`.
- Else if method == "PAYPAL" → call `payPalSubsystem.process(order)`.
- Wait for callback/confirmation if required.
- Return true if payment succeeds.

2. refundTransaction

- Parameter:

- transaction: `PaymentTransaction`

- Exception:

- `RefundNotAllowedException` if refund conditions unmet.

- Method:

- Identify method used.
- Delegate to subsystem:
 - QR refunds → VietQR
 - PayPal refunds → PayPal
- Return true if refund successful.